Worksheet 15 · Recurring Decimals IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Recognise recurring decimal notation • Convert a recurring decimal to a fraction • Prove a recurring decimal equals a given fraction algebraically		
Grade: / 19	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Recurring Decimals*

Question 1 (2 marks)

Use algebra to show that the recurring decimal $0.2\dot{5}\dot{4} = \frac{14}{55}$

Question 2 (2 marks)

Use algebra to show that the recurring decimal $0.2\dot{6} = \frac{4}{15}$

Question 3 (2 marks)

Use algebra to show that the recurring decimal $0.28\dot{1}\dot{3} = \frac{557}{1980}$

Question 4 (2 marks)

Use algebra to show that $0.3\dot{4}\dot{5} = \frac{19}{55}$

Question 5 (2 marks)

(a) Use algebra to show that $0.3\dot{2}\dot{4} = \frac{107}{330}$

(2)

Question 6 (2 marks)

Show that the recurring decimal $0.\dot{3}9\dot{6} = \frac{44}{111}$

Question 7 (3 marks)

$0.4\dot{x}$ is a recurring decimal.
x is a whole number such that $1 \leqslant x \leqslant 9$

Find, in terms of x, the recurring decimal $0.4\dot{x}$ as a fraction.
Give your fraction in its simplest form.
Show clear algebraic working.

Question 8 (2 marks)

Use algebra to show that $0.68\dot{1}\dot{1} = \frac{15}{22}$

Question 9 **(2 marks)**

Use algebra to show that $4.\dot{5}\dot{7} = 4\frac{19}{33}$